

Advanced Java Programming (3160707)

Summer – 2025 – Solution

Q.1 (a) Discuss the use of execute(), executeUpdate() and executeQuery() methods.

[3 Marks]

Answer:

These methods are used in JDBC to execute SQL queries from Java programs.

1) executeQuery()

- This method is used to execute **SELECT** queries.
- It returns data in the form of a **ResultSet** object.

Syntax:

```
ResultSet rs = stmt.executeQuery("SELECT * FROM Student");
```

Use:

Used when we want to fetch records from database.

2) executeUpdate()

- This method is used for **INSERT, UPDATE, DELETE** queries.
- It returns an integer value showing number of affected rows.

Syntax:

```
int n = stmt.executeUpdate("INSERT INTO Student VALUES(1,'Rahul')");
```

Use:

Used when database records are modified.

3) execute()

- This method can execute any type of SQL query.
- It returns:
 - **True** → if ResultSet is returned
 - **False** → if update count is returned

Syntax:

```
boolean b = stmt.execute("SELECT * FROM Student");
```

Use:

Used when query type is unknown.

Q.1 (b) Explain the following classes with their use.

[4 Marks]

Answer:

1) **URLConnection Class**

- URLConnection class is used to establish communication between application and URL resource.
- It is available in java.net package.

Use:

- Reading data from web page
- Sending data over internet

Example:

```
URL u = new URL("https://example.com");  
URLConnection con = u.openConnection();
```

2) **DatagramSocket Class**

- DatagramSocket class is used for sending and receiving UDP packets.
- It supports connectionless communication.

Use:

- Online games
- Video streaming
- Fast communication

Example:

```
DatagramSocket ds = new DatagramSocket();
```

3) **DatagramPacket Class**

- DatagramPacket class represents data packets used in UDP communication.

Use:

- Store data to send or receive through DatagramSocket.

Example:

```
byte b[] = "Hello".getBytes();  
DatagramPacket dp = new DatagramPacket(b,b.length);
```

4) **ServerSocket Class**

- ServerSocket class is used to create server-side applications.
- It waits for client request.

Use:

- Socket programming
- Client-server communication

Example:

```
ServerSocket ss = new ServerSocket(5000);
```

Q.1 (c) Design a form to input details of an employee and submit the data to a servlet. Write code for servlet that will save the entered details as a new record in database table Employee with fields (EmpId, EName, Email, Age). [7 Marks]

Answer:

HTML Form (employee.html)

```
<html>
<head>
    <title>Employee Form</title>
</head>
<body>

<h2>Employee Registration Form</h2>

<form action="EmployeeServlet"
method="post">

    Employee ID :
    <input type="text" name="eid"><br><br>

    Employee Name :
    <input type="text"
name="ename"><br><br>

    Email :
    <input type="email"
name="email"><br><br>

    Age :
    <input type="text" name="age"><br><br>

    <input type="submit" value="Save">

</form>

</body>
</html>
```

Servlet Program (EmployeeServlet.java)

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import java.sql.*;

public class EmployeeServlet extends HttpServlet
{
    public void doPost(HttpServletRequest req,
                        HttpServletResponse res)
        throws ServletException, IOException
    {
        res.setContentType("text/html");

        PrintWriter out = res.getWriter();

        String id = req.getParameter("eid");
        String name = req.getParameter("ename");
        String email = req.getParameter("email");
        String age = req.getParameter("age");

        try
        {
            // Load Driver
            Class.forName("com.mysql.jdbc.Driver");

            // Database Connection
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/test",
                "root",
                "");

            // SQL Query
            PreparedStatement ps = con.prepareStatement(
                "insert into Employee values(?,?,?,?)");

            ps.setString(1, id);
            ps.setString(2, name);
            ps.setString(3, email);
            ps.setString(4, age);
```

```
        int i = ps.executeUpdate();

        if(i > 0)
        {
            out.println("Record Inserted Successfully");
        }
        else
        {
            out.println("Record Not Inserted");
        }

        con.close();
    }
    catch(Exception e)
    {
        out.println(e);
    }
}
```

Database Table

```
CREATE TABLE Employee
(
    EmpId INT,
    EName VARCHAR(50),
    Email VARCHAR(50),
    Age INT
);
```

Explanation

- HTML form accepts employee details.
- Data is sent to servlet using POST method.
- Servlet receives data using `getParameter()`.
- JDBC is used to connect database.
- `PreparedStatement` inserts record into Employee table.

Output

Employee details are stored successfully in database table.